

FLYING QUALITIES ANALYSIS - Aircraft Model E2

Standard: MIL-F-8785C | Class III, Category B

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OVERVIEW

This script performs the complete dynamic stability and flying quality analysis of aircraft Model E2, a large high-speed aircraft ($M = 0.9$) cruising at 10 668 m altitude.

Sections:

1. Aircraft data
2. Longitudinal and lateral state / control matrices
3. Characteristic equations
4. Eigenvalues and eigenvectors
5. Stability mode characteristics
6. Time-domain simulations
7. Transfer functions + Bode diagrams + Root loci
8. MIL-F-8785C flying quality qualification
9. Stability Augmentation System (SAS) design

```
clear; clc; close all;
```

1. AIRCRAFT DATA

Geometry

```
S_E2 = 49.24; % Wing surface [m^2]
c_barre_E2 = 4.88; % Mean aerodynamic chord [m]
b_E2 = 11.80; % Wingspan [m]
A_E2 = 2.83; % Aspect ratio
e_E2 = 0.94; % Oswald efficiency number
g = 9.81; % Gravitational acceleration [m/s^2]
```

Flight condition

```
altitude_E2 = 10668; % Altitude [m]
mach_E2 = 0.9; % Mach number
U0_E2 = 267.0; % True airspeed [m/s]
q_barre_E2 = 13582; % Dynamic pressure [Pa]
theta0_E2 = 2.6; % Steady-state pitch angle [deg]
```

```
rho_E2 = 0.381; % Air density [kg/m^3]
```

Mass and inertia

```
m_E2 = 17706; % Mass [kg]  
IXX_E2 = 33910; % Roll moment of inertia [kg.m^2]  
IYY_E2 = 166157; % Pitch moment of inertia [kg.m^2]  
IZZ_E2 = 189623; % Yaw moment of inertia [kg.m^2]  
IXZ_E2 = 2984; % Product of inertia [kg.m^2]
```

Steady-state aerodynamic coefficients

```
C_L0_E2 = 0.26; % Lift coefficient at trim  
C_D0_E2 = 0.03; % Drag coefficient at trim
```

Longitudinal stability derivatives

```
C_Du_E2 = 0.027;  
C_Dalpha_E2 = 0.3;  
C_Lu_E2 = 0.27;  
C_Lalpha_E2 = 3.75;  
C_Lalphap_E2 = 0;  
C_Lq_E2 = 0;  
C_mu_E2 = 0.117;  
C_malpha_E2 = -0.4;  
C_malphap_E2 = -1.3;  
C_mq_E2 = -2.7;
```

Lateral stability derivatives

```
C_Lbeta_E2 = -0.08;  
C_Lp_E2 = -0.24;  
C_Lr_E2 = 0.07;  
C_nbeta_E2 = 0.125;  
C_nP_E2 = -0.036;  
C_nr_E2 = -0.32;  
C_ybeta_E2 = -0.68;  
C_yp_E2 = 0;  
C_yr_E2 = 0;
```

Longitudinal control derivatives

```
CD_deltae_E2 = -0.1;  
CL_deltae_E2 = 0.4;  
Cm_deltae_E2 = -0.58;
```

Lateral control derivatives

```
CY_deltaA_E2 = -0.016;  
Cl_deltaA_E2 = 0.042;  
Cn_deltaA_E2 = -0.001;  
Cy_deltar_E2 = 0.095;  
Cl_deltar_E2 = 0.006;  
Cn_deltar_E2 = -0.066;
```

2. STATE AND CONTROL MATRICES

```
fprintf('2. STATE AND CONTROL MATRICES\n');
```

2. STATE AND CONTROL MATRICES

Dimensional longitudinal stability derivatives

```
Xu_E2 = -(q_barre_E2*S_E2)/(m_E2*U0_E2) * (2*C_D0_E2 + C_Du_E2);
Xw_E2 = (q_barre_E2*S_E2)/(m_E2*U0_E2) * (C_L0_E2*(1 - (2*C_Lalpha_E2)/
(pi*e_E2*A_E2)));
Zu_E2 = -(q_barre_E2*S_E2)/(m_E2*U0_E2) * (2*C_L0_E2 + C_Lu_E2);
Zw_E2 = -(q_barre_E2*S_E2)/(m_E2*U0_E2) * (C_D0_E2 + C_Lalpha_E2);
Mu_E2 = (q_barre_E2*S_E2*c_barre_E2)/(IYY_E2*U0_E2) * C_mu_E2;
Mwp_E2 = (q_barre_E2*S_E2*(c_barre_E2^2))/(2*IYY_E2*(U0_E2^2)) * C_malphap_E2;
Mw_E2 = (q_barre_E2*S_E2*c_barre_E2)/(IYY_E2*U0_E2) * C_malpha_E2;
Mq_E2 = (q_barre_E2*S_E2*(c_barre_E2^2)*C_mq_E2)/(2*IYY_E2*U0_E2);
```

Longitudinal control derivatives

```
X_deltae_E2 = -(q_barre_E2*S_E2*CD_deltae_E2)/m_E2;
X_deltaT_E2 = 0;
Z_deltae_E2 = -(q_barre_E2*S_E2*CL_deltae_E2)/(m_E2*U0_E2);
Z_deltaT_E2 = 0;
M_deltae_E2 = (q_barre_E2*S_E2*c_barre_E2*Cm_deltae_E2)/IYY_E2;
M_deltaT_E2 = 0;
```

Longitudinal state matrix A (4x4) — states: [Du, Dw, Dq, Dtheta]

```
A_Long_E2 = [Xu_E2, Xw_E2, 0, -g*cos(theta0_E2*pi/180);
             Zu_E2, Zw_E2, U0_E2, -g*sin(theta0_E2*pi/180);
             Mu_E2 + Zu_E2*Mwp_E2, Mw_E2 + Zw_E2*Mwp_E2, Mq_E2 + U0_E2*Mwp_E2, 0;
             0, 0, 1, 0];
```

Longitudinal control matrix B (4x2) — inputs: [deltae, deltaT]

```
B_Long_E2 = [X_deltae_E2, X_deltaT_E2;
             Z_deltae_E2, Z_deltaT_E2;
             M_deltae_E2 + Z_deltae_E2*Mwp_E2, M_deltaT_E2 + Z_deltaT_E2*Mwp_E2;
             0, 0];
```

```
fprintf('Longitudinal State Matrix A_Long:\n');
```

Longitudinal State Matrix A_Long:

```
disp(A_Long_E2)
```

```
-0.0123    0.0038         0    -9.7999
-0.1118   -0.5347   267.0000   -0.4450
 0.0087   -0.0290   -0.7180         0
         0         0    1.0000         0
```

Row 1 (Du eq.): $X_u = -0.0123$ (speed drag damping), $X_w = +0.0038$ (small lift coupling)

Row 2 (Dw eq.): $Z_u = -0.1118$ (speed-lift coupling), $Z_w = -0.5347$ (heave damping), $U_0 = 267$ (centripetal)

Row 3 (Dq eq.): pitch equation; $M_q + U_0 * \dot{M}_w = -0.718$ gives the pitch damping

Row 4 (Dtheta): kinematic relation $d\theta / dt = q \rightarrow$ coefficient 1 on Dq

The large U_0 term (267) in row 2 couples angle of attack to pitch rate.

```
fprintf('Longitudinal Control Matrix B_Long:\n');
```

Longitudinal Control Matrix B_Long:

```
disp(B_Long_E2)
```

```
 3.7771         0
-0.0566         0
-11.3922        0
         0         0
```

$X_{\delta a} = 3.78 \text{ m/s}^2/\text{rad}$: elevator creates a forward force (drag term negative $\rightarrow$ positive X_u)

$Z_{\delta a} = -0.057$: small normal force from elevator

$M_{\delta a} = -11.39 \text{ rad/s}^2/\text{rad}$: elevator is a powerful pitch moment generator (nose-down negative)

The thrust column (col 2) is zero throughout, throttle decoupled from pitch in this model.

Dimensional lateral stability derivatives

```
Yv_E2 = (q_barre_E2*S_E2*C_ybeta_E2)/(m_E2*U0_E2);
Yp_E2 = (q_barre_E2*S_E2*C_yp_E2)/(2*m_E2*U0_E2);
Yr_E2 = (q_barre_E2*S_E2*C_yr_E2)/(2*m_E2*U0_E2);
Lv_E2 = (q_barre_E2*S_E2*b_E2*C_Lbeta_E2)/(IXX_E2*U0_E2);
Lp_E2 = (q_barre_E2*S_E2*(b_E2^2)*C_Lp_E2)/(2*IXX_E2*U0_E2);
Lr_E2 = (q_barre_E2*S_E2*(b_E2^2)*C_Lr_E2)/(2*IXX_E2*U0_E2);
Nv_E2 = (q_barre_E2*S_E2*b_E2*C_nbeta_E2)/(IZZ_E2*U0_E2);
Np_E2 = (q_barre_E2*S_E2*(b_E2^2)*C_nP_E2)/(2*IZZ_E2*U0_E2);
Nr_E2 = (q_barre_E2*S_E2*(b_E2^2)*C_nr_E2)/(2*IZZ_E2*U0_E2);
```

Lateral control derivatives

```
Y_deltar_E2 = (q_barre_E2*S_E2*Cy_deltar_E2)/(m_E2*U0_E2);
Y_deltaa_E2 = (q_barre_E2*S_E2*CY_deltaA_E2)/(m_E2*U0_E2);
L_deltar_E2 = (q_barre_E2*S_E2*b_E2*Cl_deltar_E2)/(IXX_E2*U0_E2);
L_deltaa_E2 = (q_barre_E2*S_E2*b_E2*Cl_deltaA_E2)/(IXX_E2*U0_E2);
N_deltar_E2 = (q_barre_E2*S_E2*b_E2*Cn_deltar_E2)/(IZZ_E2*U0_E2);
```

```
N_deltaa_E2 = (q_barre_E2*S_E2*b_E2*Cn_deltaA_E2)/(IZZ_E2*U0_E2);
```

Lateral state matrix A (4x4) — states: [Dv, Dp, Dr, Dphi]

```
A_lat_E2 = [Yv_E2, Yp_E2, (Yr_E2 - U0_E2), g*cos(theta0_E2*pi/180);
             Lv_E2, Lp_E2, Lr_E2, 0;
             Nv_E2, Np_E2, Nr_E2, 0;
             0, 1, 0, 0];
```

Lateral control matrix B (4x2) — inputs: [deltar, deltaa]

```
B_lat_E2 = [Y_deltar_E2, Y_deltaa_E2;
             L_deltar_E2, L_deltaa_E2;
             N_deltar_E2, N_deltaa_E2;
             0, 0];
```

```
fprintf('Lateral State Matrix A_lat:\n');
```

Lateral State Matrix A_lat:

```
disp(A_lat_E2)
```

```
-0.0962      0 -267.0000    9.7999
-0.0697 -1.2342    0.3600      0
 0.0195 -0.0331 -0.2943      0
      0    1.0000      0      0
```

Yv = -0.096: sideslip produces a restoring lateral force (side force damping)

(Yr-U0) = -267: the large -U0 term comes from centripetal acceleration in the Dv equation

Lp = -1.234 : strong roll damping: roll rate is quickly attenuated

Nr = -0.294 : yaw damping: positive yaw rate is damped

g*cos(theta0) = 9.80 in row 1 col 4: gravity couples bank angle to sideslip (coordinated turn)

```
fprintf('Lateral Control Matrix B_lat:\n');
```

Lateral Control Matrix B_lat:

```
disp(B_lat_E2)
```

```
 0.0134 -0.0023
 0.0052  0.0366
-0.0103 -0.0002
      0      0
```

Col 1 (rudder): N_deltar = -0.010 gives yaw moment, L_deltar = +0.005 gives small roll coupling

Col 2 (aileron): L_deltaa = +0.037 is the main roll moment, Cn_deltaa = -0.0002 (adverse yaw, small)

3. CHARACTERISTIC EQUATIONS

```
fprintf('3. CHARACTERISTIC EQUATIONS\n');
```

3. CHARACTERISTIC EQUATIONS

```
syms lambda
```

```
Char_Long_E2 = det(lambda*eye(4) - A_Long_E2);  
Char_Long_E2 = expand(Char_Long_E2);  
Char_Long_vpa_E2 = vpa(Char_Long_E2, 6);
```

```
Char_Lat_E2 = det(lambda*eye(4) - A_lat_E2);  
Char_Lat_E2 = expand(Char_Lat_E2);  
Char_Lat_vpa_E2 = vpa(Char_Lat_E2, 6);
```

```
fprintf('Longitudinal Characteristic Polynomial:\n');
```

Longitudinal Characteristic Polynomial:

```
disp(Char_Long_vpa_E2)
```

$$\lambda^4 + 1.26504 \lambda^3 + 8.13175 \lambda^2 + 0.163839 \lambda + 0.0771884$$

$$\lambda^4 + 1.265\lambda^3 + 8.132\lambda^2 + 0.1638\lambda + 0.0772 = 0$$

- Degree 4 $\rightarrow$ 4 eigenvalues $\rightarrow$ 2 complex conjugate pairs (short period + phugoid)
- All coefficients positive $\rightarrow$ necessary condition for stability (Routh criterion satisfied)
- The large λ^2 coefficient (8.13) reflects the high short-period natural frequency
- The very small λ^1 coefficient (0.164) and constant (0.077) are signatures of the lightly damped, slow phugoid mode

```
fprintf('Lateral Characteristic Polynomial:\n');
```

Lateral Characteristic Polynomial:

```
disp(Char_Lat_vpa_E2)
```

$$\lambda^4 + 1.62469 \lambda^3 + 5.72431 \lambda^2 + 7.75632 \lambda + 0.132361$$

$$\lambda^4 + 1.625\lambda^3 + 5.724\lambda^2 + 7.756\lambda + 0.132 = 0$$

- Again degree 4 $\rightarrow$ 2 real eigenvalues (roll + spiral) + 1 complex pair (Dutch roll)
- All coefficients positive $\rightarrow$ lateral system is stable
- The λ^1 coefficient (7.756) is relatively large, driven by the strong directional restoring moment (N_v term) that governs Dutch roll
- The near-zero constant (0.132) indicates that the spiral mode eigenvalue is very close to zero (very slow spiral dynamics)

4. EIGENVALUES AND EIGENVECTORS

```
fprintf('4. EIGENVALUES AND EIGENVECTORS\n');
```

4. EIGENVALUES AND EIGENVECTORS

Longitudinal

```
eig_Long_E2 = eig(A_Long_E2);  
[V_Long_E2, D_Long_E2] = eig(A_Long_E2);  
  
fprintf('Longitudinal Eigenvalues:\n');
```

Longitudinal Eigenvalues:

```
disp(eig_Long_E2)
```

```
-0.6231 + 2.7768i  
-0.6231 - 2.7768i  
-0.0094 + 0.0972i  
-0.0094 - 0.0972i
```

$-0.6231 \pm 2.7768i \rightarrow$ SHORT PERIOD mode

Real part negative $\rightarrow$ damped oscillation.

Large imaginary part (2.78 rad/s) $\rightarrow$ fast oscillation (~ 2.3 s period).

This is the dominant mode felt by the pilot during pitch manoeuvres.

$-0.0094 \pm 0.0972i \rightarrow$ PHUGOID mode

Very small real part $\rightarrow$ very lightly damped ($\zeta \approx 0.083$).

Very small imaginary part $\rightarrow$ very slow oscillation (~ 99 s period).

This is the long-period energy exchange between kinetic and potential energy.

The pilot has ample time to correct it.

```
fprintf('Longitudinal Eigenvectors (columns correspond to eigenvalues  
above):\n');
```

Longitudinal Eigenvectors (columns correspond to eigenvalues above):

```
disp(V_Long_E2)
```

```
-0.0047 - 0.0103i  -0.0047 + 0.0103i   0.9636 + 0.0000i   0.9636 + 0.0000i  
-0.9999 + 0.0000i  -0.9999 + 0.0000i   0.2671 - 0.0049i   0.2671 + 0.0049i  
 0.0003 - 0.0104i   0.0003 + 0.0104i   0.0009 + 0.0001i   0.0009 - 0.0001i  
-0.0036 + 0.0007i  -0.0036 - 0.0007i  -0.0002 - 0.0096i  -0.0002 + 0.0096i
```

Col 1-2 (short period): dominant component in rows 1 (Du) $\rightarrow$ actually row 2 (Dw = -0.9999)

The angle of attack perturbation Dw dominates the short period $\rightarrow$ pilot feels pitch.

Col 3-4 (phugoid): dominant component in row 1 ($Du = 0.964$) and row 4 ($D\theta = 0.010i$)

Speed Du and pitch angle $D\theta$ dominate the phugoid $\rightarrow$ classic energy exchange.

Lateral

```
eig_Lat_E2 = eig(A_lat_E2);  
[V_Lat_E2, D_Lat_E2] = eig(A_lat_E2);  
  
fprintf('Lateral Eigenvalues:\n');
```

Lateral Eigenvalues:

```
disp(eig_Lat_E2)
```

```
-0.0974 + 2.3263i  
-0.0974 - 2.3263i  
-1.4126 + 0.0000i  
-0.0173 + 0.0000i
```

$-0.0974 \pm 2.3263i \rightarrow$ DUTCH ROLL mode

Complex pair: coupled yaw-roll oscillation, period ≈ 2.76 s.

Low damping ratio ($\zeta \approx 0.085$) $\rightarrow$ many oscillation cycles before decay.

Typical of large transport-category aircraft; meets MIL Level 1 marginally.

$-1.4126 \rightarrow$ ROLL mode (real, aperiodic)

Large negative real part $\rightarrow$ very fast roll rate decay, time constant ≈ 0.71 s.

Pilot perceives immediate roll response to aileron, then exponential decay.

$-0.0173 \rightarrow$ SPIRAL mode (real, aperiodic)

Very small negative real part $\rightarrow$ very slow convergence, time constant ≈ 58 s.

Negative $\rightarrow$ stable spiral: a disturbed bank angle slowly returns to zero.

```
fprintf('Lateral Eigenvectors:\n');
```

Lateral Eigenvectors:

```
disp(V_Lat_E2)
```

```
-0.9995 + 0.0000i  -0.9995 + 0.0000i  -0.8970 + 0.0000i   0.4400 + 0.0000i  
 0.0129 - 0.0236i   0.0129 + 0.0236i  -0.3607 + 0.0000i  -0.0155 + 0.0000i  
-0.0004 + 0.0085i  -0.0004 - 0.0085i   0.0050 + 0.0000i   0.0328 + 0.0000i  
-0.0104 - 0.0051i  -0.0104 + 0.0051i   0.2554 + 0.0000i   0.8973 + 0.0000i
```

Col 1-2 (Dutch roll): all four lateral states contribute (sideslip Dv dominant at -0.9995)

Col 3 (roll mode): $Dv = -0.897$ and $Dp = -0.361$ dominate $\rightarrow$ pure roll motion

Col 4 (spiral mode): $D_{\phi} = 0.897$ dominant $\rightarrow$ slow bank angle divergence/convergence

5. STABILITY MODE CHARACTERISTICS

```
fprintf('5. STABILITY MODE CHARACTERISTICS\n');
```

5. STABILITY MODE CHARACTERISTICS

Short Period

```
A_SP_E2 = [Zw_E2, U0_E2; Mw_E2 + Zw_E2*Mwp_E2, Mq_E2 + U0_E2*Mwp_E2];  
eig_SP_E2 = eig(A_SP_E2);  
wn_SP_E2 = abs(eig_SP_E2(1));  
zeta_SP_E2 = -real(eig_SP_E2(1)) / wn_SP_E2;  
T_SP_E2 = 2*pi / (wn_SP_E2 * sqrt(1 - zeta_SP_E2^2));
```

```
fprintf('SHORT PERIOD MODE\n');
```

SHORT PERIOD MODE

```
fprintf('Eigenvalues: %.4f +/- %.4fi\n', real(eig_SP_E2(1)), imag(eig_SP_E2(1)));
```

Eigenvalues: -0.6264 +/- 2.7791i

```
fprintf('Nat. freq.: wn = %.4f rad/s\n', wn_SP_E2);
```

Nat. freq.: $wn = 2.8488$ rad/s

```
fprintf('Damping ratio: zeta = %.4f\n', zeta_SP_E2);
```

Damping ratio: $zeta = 0.2199$

```
fprintf('Period: T = %.4f s\n\n', T_SP_E2);
```

Period: $T = 2.2608$ s

$wn = 2.85$ rad/s $\rightarrow$ fast oscillation, period $T \approx 2.26$ s

$zeta = 0.2199 \rightarrow$ moderately underdamped; the aircraft oscillates in pitch but the oscillation decays within ~ 5 cycles (~ 12 s).

MIL-F-8785C Level 1 requires $zeta \in [0.30, 2.00]$ for Class III Cat B.

$\rightarrow zeta = 0.22$ FAILS Level 1 $\rightarrow$ classified Level 2 (acceptable but increased pilot workload during pitch manoeuvres).

A pitch damper (SAS) is required to bring $zeta$ above 0.30.

Phugoid

```
A_phug_E2 = [Xu_E2, -g*cos(theta0_E2*pi/180); -Zu_E2/U0_E2,  
(g*sin(theta0_E2*pi/180))/U0_E2];  
eig_phug_E2 = eig(A_phug_E2);
```

```

wn_phug_E2 = abs(eig_phug_E2(1));
zeta_phug_E2 = -real(eig_phug_E2(1)) / wn_phug_E2;
T_phug_E2 = 2*pi / (wn_phug_E2 * sqrt(max(1 - zeta_phug_E2^2, 0)));

fprintf('PHUGOID MODE\n');

```

PHUGOID MODE

```

fprintf('Eigenvalues : %.4f +/- %.4fi\n', real(eig_phug_E2(1)),
imag(eig_phug_E2(1)));

```

Eigenvalues : -0.0053 +/- 0.0637i

```

fprintf('Nat. freq. : wn = %.4f rad/s\n', wn_phug_E2);

```

Nat. freq. : wn = 0.0639 rad/s

```

fprintf('Damping ratio: zeta = %.4f\n', zeta_phug_E2);

```

Damping ratio: zeta = 0.0833

```

fprintf('Period : T = %.4f s\n\n', T_phug_E2);

```

Period : T = 98.6929 s

$\omega_n = 0.064 \text{ rad/s} \rightarrow$ very slow oscillation, period $T \approx 99 \text{ s}$ ($\sim 1.6 \text{ min}$)

$\zeta = 0.083 \rightarrow$ light damping; the oscillation slowly fades over many minutes.

MIL Level 1 requires $\zeta_{ph} \geq 0.04 \rightarrow$ SATISFIED (Level 1).

The slow period means a human pilot can easily detect and correct a phugoid excitation; it is essentially invisible during normal operation.

Lateral aperiodic modes

```

real_eigs = eig_Lat_E2(imag(eig_Lat_E2) == 0);
[~, idx_spiral] = min(abs(real(real_eigs)));
[~, idx_roll] = max(abs(real(real_eigs)));
lambda_spiral_E2 = real_eigs(idx_spiral);
lambda_roll_E2 = real_eigs(idx_roll);
T_spiral_E2 = abs(1 / real(lambda_spiral_E2));
T_roll_E2 = abs(1 / real(lambda_roll_E2));

fprintf('SPIRAL MODE\n');

```

SPIRAL MODE

```

fprintf('Eigenvalue : lambda = %.4f\n', real(lambda_spiral_E2));

```

Eigenvalue : lambda = -0.0173

```

fprintf('Time constant : T = %.4f s\n\n', T_spiral_E2);

```

Time constant : T = 57.8559 s

Spiral:

$\lambda = -0.017 < 0 \rightarrow$ stable (convergent spiral).

Time constant $T \approx 58$ s $\rightarrow$ extremely slow: a disturbed bank angle takes nearly a minute to return halfway to wings-level.

MIL Level 1 requires either stability ($\lambda < 0$) or time-to-double > 20 s.

$\rightarrow$ SATISFIED (Level 1).

```
fprintf('ROLL MODE\n');
```

ROLL MODE

```
fprintf('Eigenvalue : lambda = %.4f\n', real(lambda_roll_E2));
```

Eigenvalue : lambda = -1.4126

```
fprintf('Time constant : T = %.4f s\n\n', T_roll_E2);
```

Time constant : T = 0.7079 s

Roll mode:

$\lambda = -1.41 \rightarrow$ fast aperiodic decay.

Time constant $T \approx 0.71$ s $\rightarrow$ roll rate attenuates to 37% of its initial value in less than one second. Pilot perceives immediate, crisp roll response.

MIL Level 1 requires $T_{\text{roll}} \leq 1.4$ s $\rightarrow$ SATISFIED (Level 1).

Dutch Roll

```
A_DR_E2 = [Yv_E2, (Yr_E2 - U0_E2); Nv_E2, Nr_E2];  
eig_DR_E2 = eig(A_DR_E2);  
cplx_idx = find(imag(eig_DR_E2) ~= 0, 1);  
if ~isempty(cplx_idx)  
    wn_DR_E2 = abs(eig_DR_E2(cplx_idx));  
    zeta_DR_E2 = -real(eig_DR_E2(cplx_idx)) / wn_DR_E2;  
else  
    cplx_eig = eig_Lat_E2(imag(eig_Lat_E2) ~= 0);  
    wn_DR_E2 = abs(cplx_eig(1));  
    zeta_DR_E2 = -real(cplx_eig(1)) / wn_DR_E2;  
end  
T_DR_E2 = 2*pi / (wn_DR_E2 * sqrt(max(1 - zeta_DR_E2^2, 0)));  
  
fprintf('DUTCH ROLL MODE\n');
```

DUTCH ROLL MODE

```
fprintf('Nat. freq. : wn = %.4f rad/s\n', wn_DR_E2);
```

Nat. freq. : wn = 2.2870 rad/s

```
fprintf('Damping ratio: zeta = %.4f\n', zeta_DR_E2);
```

Damping ratio: zeta = 0.0854

```
fprintf('zeta*wn = %.4f rad/s\n', zeta_DR_E2*wn_DR_E2);
```

```
zeta*wn = 0.1952 rad/s
```

```
fprintf('Period : T = %.4f s\n\n', T_DR_E2);
```

```
Period : T = 2.7574 s
```

$\omega_n = 2.29 \text{ rad/s}$, $T \approx 2.76 \text{ s} \rightarrow$ moderate frequency coupled yaw-roll oscillation.

$\zeta = 0.085 \rightarrow$ low damping; the aircraft sways left-right for many cycles.

MIL Level 1 (Class III Cat B) requires:

$\zeta_{DR} \geq 0.08 \rightarrow 0.085 \geq 0.08 \rightarrow$ OK (barely)

$\zeta \omega_n \geq 0.15 \rightarrow 0.195 \geq 0.15 \rightarrow$ OK

$\omega_{DR} \geq 0.4 \text{ rad/s} \rightarrow 2.29 \geq 0.40 \rightarrow$ OK

$\rightarrow$ SATISFIED (Level 1) but only marginally on the damping requirement.

A yaw damper could improve the robustness of this margin.

6. TIME-DOMAIN SIMULATIONS

```
fprintf('6. TIME-DOMAIN SIMULATIONS\n');
```

```
6. TIME-DOMAIN SIMULATIONS
```

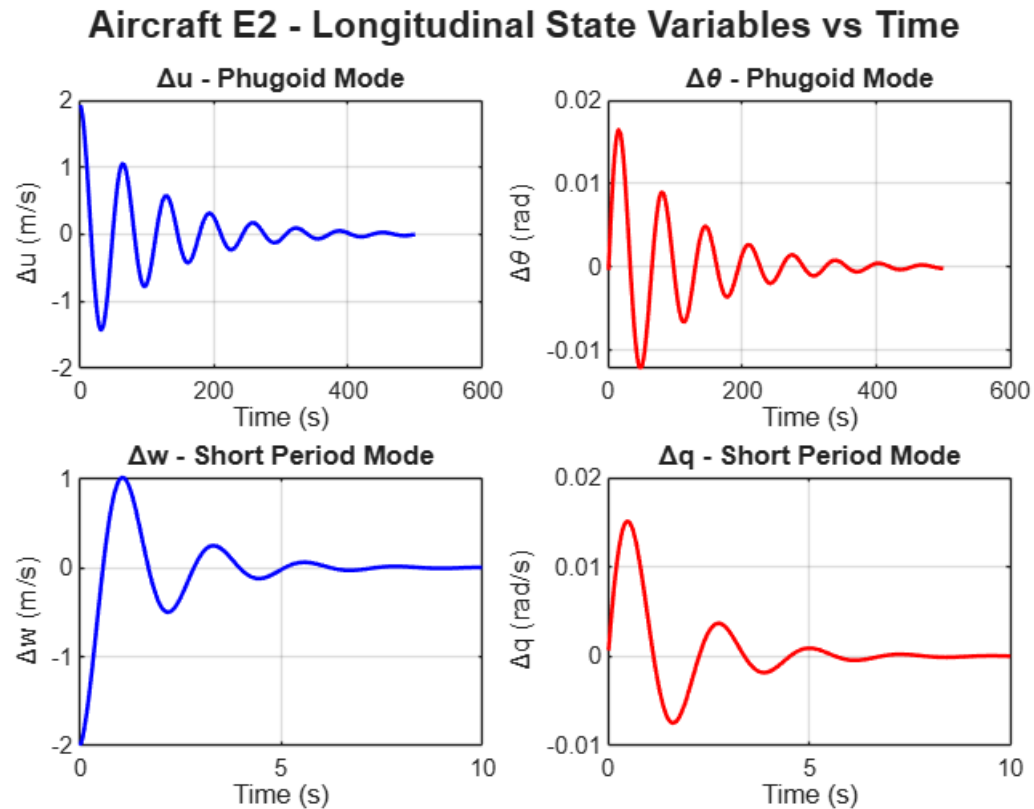
Longitudinal modes

```
t_long = 0:0.01:500;  
t_sp = 0:0.01:10;  
  
cplx_long = eig_Long_E2(imag(eig_Long_E2) > 0);  
[~, srt] = sort(abs(cplx_long), 'descend');  
lambda_SP1 = cplx_long(srt(1)); lambda_SP2 = conj(lambda_SP1);  
lambda_ph1 = cplx_long(srt(2)); lambda_ph2 = conj(lambda_ph1);  
  
Du_phug = real(V_Long_E2(1,3)*exp(lambda_ph1*t_long) +  
V_Long_E2(1,4)*exp(lambda_ph2*t_long));  
Dtheta_phug = real(V_Long_E2(4,3)*exp(lambda_ph1*t_long) +  
V_Long_E2(4,4)*exp(lambda_ph2*t_long));  
Dw_sp = real(V_Long_E2(2,1)*exp(lambda_SP1*t_sp) +  
V_Long_E2(2,2)*exp(lambda_SP2*t_sp));  
Dq_sp = real(V_Long_E2(3,1)*exp(lambda_SP1*t_sp) +  
V_Long_E2(3,2)*exp(lambda_SP2*t_sp));  
  
figure('Name','Longitudinal Modes - E2','Position',[100 100 1000 700]);  
subplot(2,2,1); plot(t_long, Du_phug, 'b', 'LineWidth', 1.5);  
title('\Delta u - Phugoid Mode'); xlabel('Time (s)'); ylabel('\Delta u (m/s)');  
grid on;
```

```

subplot(2,2,2); plot(t_long, Dtheta_phug, 'r', 'LineWidth', 1.5);
title('\Delta\theta - Phugoid Mode'); xlabel('Time (s)'); ylabel('\Delta\theta (rad)'); grid on;
subplot(2,2,3); plot(t_sp, Dw_sp, 'b', 'LineWidth', 1.5);
title('\Deltaw - Short Period Mode'); xlabel('Time (s)'); ylabel('\Deltaw (m/s)'); grid on;
subplot(2,2,4); plot(t_sp, Dq_sp, 'r', 'LineWidth', 1.5);
title('\Deltaq - Short Period Mode'); xlabel('Time (s)'); ylabel('\Deltaq (rad/s)'); grid on;
sgtitle('Aircraft E2 - Longitudinal State Variables vs Time', 'FontWeight',
'bold');
saveas(gcf, 'fig_longitudinal_modes.png');

```



Top row (phugoid, 500 s window):

Δu and $\Delta \theta$ oscillate with a period of ~ 99 s, slowly decaying.

The light damping ($\zeta=0.083$) means ~ 6 full oscillation cycles are still visible before the amplitude drops significantly.

This confirms the phugoid is not a handling concern for the pilot.

Bottom row (short period, 10 s window):

Δw and Δq oscillate rapidly ($T \approx 2.3$ s) and decay within ~ 8 s.

The damping $\zeta=0.22$ is clearly visible: the oscillation is noticeably underdamped (≈ 3 cycles before decay), which is why this mode fails Level 1.

Lateral modes

```

t_dr = 0:0.01:60;
t_roll = 0:0.01:5;
t_spiral = 0:1:500;

cplx_lat = eig_Lat_E2(imag(eig_Lat_E2) > 0);
lambda_DR1 = cplx_lat(1); lambda_DR2 = conj(lambda_DR1);

real_eig_idx = find(imag(eig_Lat_E2) == 0);
[~, i_roll] = max(abs(real(eig_Lat_E2(real_eig_idx))));
[~, i_spiral] = min(abs(real(eig_Lat_E2(real_eig_idx))));
col_roll = real_eig_idx(i_roll);
col_spiral = real_eig_idx(i_spiral);
cplx_cols = find(imag(eig_Lat_E2) > 0);
col_DR1 = cplx_cols(1);
col_DR2 = col_DR1 + 1;

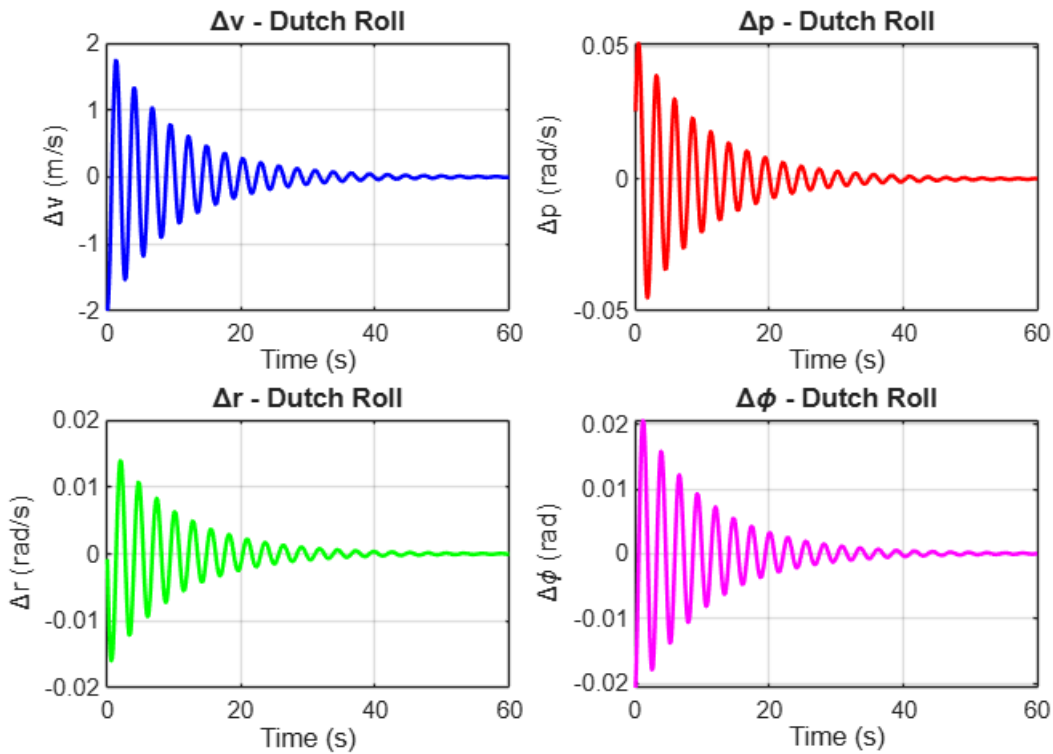
Dv_dr = real(V_Lat_E2(1,col_DR1)*exp(lambda_DR1*t_dr) +
V_Lat_E2(1,col_DR2)*exp(lambda_DR2*t_dr));
Dp_dr = real(V_Lat_E2(2,col_DR1)*exp(lambda_DR1*t_dr) +
V_Lat_E2(2,col_DR2)*exp(lambda_DR2*t_dr));
Dr_dr = real(V_Lat_E2(3,col_DR1)*exp(lambda_DR1*t_dr) +
V_Lat_E2(3,col_DR2)*exp(lambda_DR2*t_dr));
Dphi_dr = real(V_Lat_E2(4,col_DR1)*exp(lambda_DR1*t_dr) +
V_Lat_E2(4,col_DR2)*exp(lambda_DR2*t_dr));

Dp_roll = real(V_Lat_E2(2,col_roll) * exp(lambda_roll_E2 * t_roll));
Dphi_roll = real(V_Lat_E2(4,col_roll) * exp(lambda_roll_E2 * t_roll));
Dphi_spiral = real(V_Lat_E2(4,col_spiral) * exp(lambda_spiral_E2 * t_spiral));
Dr_spiral = real(V_Lat_E2(3,col_spiral) * exp(lambda_spiral_E2 * t_spiral));

figure('Name','Dutch Roll Mode - E2','Position',[100 100 1000 700]);
subplot(2,2,1); plot(t_dr, Dv_dr, 'b', 'LineWidth', 1.5);
title('\Deltav - Dutch Roll'); xlabel('Time (s)'); ylabel('\Deltav (m/s)'); grid
on;
subplot(2,2,2); plot(t_dr, Dp_dr, 'r', 'LineWidth', 1.5);
title('\Deltap - Dutch Roll'); xlabel('Time (s)'); ylabel('\Deltap (rad/s)');
grid on;
subplot(2,2,3); plot(t_dr, Dr_dr, 'g', 'LineWidth', 1.5);
title('\Deltar - Dutch Roll'); xlabel('Time (s)'); ylabel('\Deltar (rad/s)');
grid on;
subplot(2,2,4); plot(t_dr, Dphi_dr, 'm', 'LineWidth', 1.5);
title('\Delta\phi - Dutch Roll'); xlabel('Time (s)'); ylabel('\Delta\phi
(rad)'); grid on;
sgtitle('Aircraft E2 - Dutch Roll Mode', 'FontWeight', 'bold');
saveas(gcf, 'fig_dutch_roll.png');

```

Aircraft E2 - Dutch Roll Mode



All four lateral states oscillate simultaneously with period $T \approx 2.76$ s.

The low damping ($\zeta = 0.085$) means the yaw-roll oscillation persists for many cycles — visible as ~20 full swings before the envelope fades.

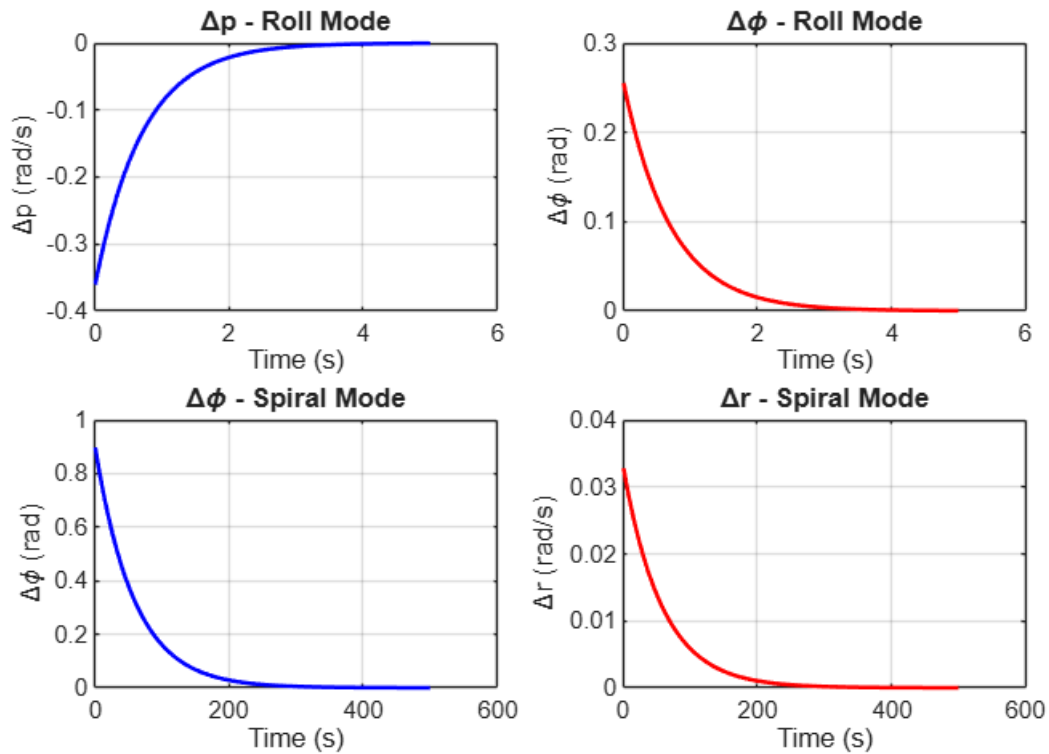
Δv (sideslip) shows the largest amplitude, confirming sideslip dominates Dutch roll.

Δp and Δr are smaller but in phase opposition, illustrating the coupled yaw-roll motion.

This oscillation, although Level 1 compliant, would be noticeable to passengers and uncomfortable on a transport aircraft, hence the interest in a yaw damper.

```
figure('Name','Roll and Spiral Modes - E2','Position',[100 100 1000 700]);
subplot(2,2,1); plot(t_roll, Dp_roll, 'b', 'LineWidth', 1.5);
title('\Delta p - Roll Mode'); xlabel('Time (s)'); ylabel('\Delta p (rad/s)');
grid on;
subplot(2,2,2); plot(t_roll, Dphi_roll, 'r', 'LineWidth', 1.5);
title('\Delta\phi - Roll Mode'); xlabel('Time (s)'); ylabel('\Delta\phi (rad)');
grid on;
subplot(2,2,3); plot(t_spiral, Dphi_spiral, 'b', 'LineWidth', 1.5);
title('\Delta\phi - Spiral Mode'); xlabel('Time (s)'); ylabel('\Delta\phi (rad)');
grid on;
subplot(2,2,4); plot(t_spiral, Dr_spiral, 'r', 'LineWidth', 1.5);
title('\Delta r - Spiral Mode'); xlabel('Time (s)'); ylabel('\Delta r (rad/s)');
grid on;
sgtitle('Aircraft E2 - Roll and Spiral Modes', 'FontWeight', 'bold');
saveas(gcf, 'fig_roll_spiral.png');
```

Aircraft E2 - Roll and Spiral Modes



Top row (roll mode, 5 s window):

Δp decays exponentially to zero in ~ 3 s ($T_{\text{roll}} \approx 0.71$ s).

$\Delta \phi$ drops quickly: an initial bank angle perturbation disappears rapidly.

This fast, clean roll response is desirable and meets Level 1.

Bottom row (spiral mode, 500 s window):

$\Delta \phi$ decays extremely slowly from an initial perturbation.

The time constant $T \approx 58$ s means a 1° bank disturbance takes ~ 4 minutes to halve.

This is a stable spiral: the aircraft slowly self-corrects without pilot input.

The yaw rate Δr follows the same slow convergence.

7. TRANSFER FUNCTIONS, BODE DIAGRAMS AND ROOT LOCUS

```
fprintf('7. TRANSFER FUNCTIONS + BODE + ROOT LOCUS\n');
```

```
7. TRANSFER FUNCTIONS + BODE + ROOT LOCUS
```

```
s = tf('s');
A = A_Long_E2;
B = B_Long_E2;
```



```

A_lat = A_lat_E2;
B_lat = B_lat_E2;

C_u = [1 0 0 0];
C_w = [0 1 0 0];
C_q = [0 0 1 0];
C_theta = [0 0 0 1];
C_v = [1 0 0 0];
C_p = [0 1 0 0];
C_r = [0 0 1 0];
C_phi = [0 0 0 1];

```

Longitudinal TFs (elevator input)

```

G_u_deltae = tf(ss(A, B(:,1), C_u, 0));
G_w_deltae = tf(ss(A, B(:,1), C_w, 0));
G_q_deltae = tf(ss(A, B(:,1), C_q, 0));
G_theta_deltae = tf(ss(A, B(:,1), C_theta, 0));

fprintf('Longitudinal Transfer Functions (elevator input)\n');

```

Longitudinal Transfer Functions (elevator input)

```
fprintf('G_u/deltae:\n'); disp(G_u_deltae)
```

```

G_u/deltae:
  tf with properties:

    Numerator: {[0 3.7771 4.7315 130.8214 59.6542]}
  Denominator: {[1 1.2650 8.1317 0.1638 0.0772]}
    Variable: 's'
    IODelay: 0
    InputDelay: 0
    OutputDelay: 0
    InputName: {''}
    InputUnit: {''}
    InputGroup: [1x1 struct]
    OutputName: {''}
    OutputUnit: {''}
    OutputGroup: [1x1 struct]
    Notes: [0x1 string]
    UserData: []
    Name: ''
    Ts: 0
    TimeUnit: 'seconds'
    SamplingGrid: [1x1 struct]

```

DC gain = 772.8 m/s per rad. A 1 rad elevator step ultimately produces a ~773 m/s speed change which is a very large static gain reflecting the slow phugoid response. The two resonant peaks correspond to $\omega_{n_SP} = 2.85$ and $\omega_{n_ph} = 0.064$ rad/s visible in the Bode magnitude plot.

```
fprintf('G_w/deltae:\n'); disp(G_w_deltae)
```

```

G_w/deltae:
  tf with properties:

    Numerator: {[0 -0.0566 -3.0422e+03 -23.8912 -12.4340]}
  Denominator: {[1 1.2650 8.1317 0.1638 0.0772]}
    Variable: 's'

```

```

        IODelay: 0
        InputDelay: 0
        OutputDelay: 0
        InputName: {''}
        InputUnit: {''}
        InputGroup: [1x1 struct]
        OutputName: {''}
        OutputUnit: {''}
        OutputGroup: [1x1 struct]
        Notes: [0x1 string]
        UserData: []
        Name: ''
        Ts: 0
        TimeUnit: 'seconds'
        SamplingGrid: [1x1 struct]

```

Large negative coefficient (~ -3042) in numerator due to the direct Zw coupling. DC gain = -161 m/s per rad.

```
fprintf('G_q/deltae:\n'); disp(G_q_deltae)
```

G_q/deltae:

tf with properties:

```

        Numerator: {[0 -11.3922 -6.1976 -0.0500 9.3867e-21]}
        Denominator: {[1 1.2650 8.1317 0.1638 0.0772]}
        Variable: 's'
        IODelay: 0
        InputDelay: 0
        OutputDelay: 0
        InputName: {''}
        InputUnit: {''}
        InputGroup: [1x1 struct]
        OutputName: {''}
        OutputUnit: {''}
        OutputGroup: [1x1 struct]
        Notes: [0x1 string]
        UserData: []
        Name: ''
        Ts: 0
        TimeUnit: 'seconds'
        SamplingGrid: [1x1 struct]

```

Numerator has a near-zero constant ($9.4e-21 \approx 0$) $\rightarrow$ G_q has a zero at $s = 0$, meaning the elevator produces no steady-state pitch rate in trim (as expected).

```
fprintf('G_theta/deltae:\n'); disp(G_theta_deltae)
```

G_theta/deltae:

tf with properties:

```

        Numerator: {[0 0 -11.3922 -6.1976 -0.0500]}
        Denominator: {[1 1.2650 8.1317 0.1638 0.0772]}
        Variable: 's'
        IODelay: 0
        InputDelay: 0
        OutputDelay: 0
        InputName: {''}
        InputUnit: {''}
        InputGroup: [1x1 struct]
        OutputName: {''}
        OutputUnit: {''}
        OutputGroup: [1x1 struct]

```

```

Notes: [0x1 string]
UserData: []
Name: ''
Ts: 0
TimeUnit: 'seconds'
SamplingGrid: [1x1 struct]

```

DC gain = -0.647 rad/rad. Negative because a positive (trailing-edge down) elevator deflection pitches the nose down. Zeros at $s=-0.536$ and $s=-0.008$ (slow zero near phugoid frequency).

Lateral TFs

```

G_v_deltar = tf(ss(A_lat, B_lat(:,1), C_v, 0));
G_p_deltar = tf(ss(A_lat, B_lat(:,1), C_p, 0));
G_r_deltar = tf(ss(A_lat, B_lat(:,1), C_r, 0));
G_phi_deltar = tf(ss(A_lat, B_lat(:,1), C_phi, 0));
G_v_deltaa = tf(ss(A_lat, B_lat(:,2), C_v, 0));
G_p_deltaa = tf(ss(A_lat, B_lat(:,2), C_p, 0));
G_r_deltaa = tf(ss(A_lat, B_lat(:,2), C_r, 0));
G_phi_deltaa = tf(ss(A_lat, B_lat(:,2), C_phi, 0));

fprintf('G_phi/deltaa:\n'); disp(G_phi_deltaa)

```

```

G_phi/deltaa:
tf with properties:

    Numerator: {[0 0 0.0366 0.0144 0.1886]}
  Denominator: {[1 1.6247 5.7243 7.7563 0.1324]}
    Variable: 's'
      IODelay: 0
    InputDelay: 0
   OutputDelay: 0
    InputName: {''}
   InputUnit: {''}
   InputGroup: [1x1 struct]
   OutputName: {''}
  OutputUnit: {''}
  OutputGroup: [1x1 struct]
        Notes: [0x1 string]
      UserData: []
        Name: ''
          Ts: 0
    TimeUnit: 'seconds'
  SamplingGrid: [1x1 struct]

```

DC gain = 1.425 rad/rad. Zeros at $-0.197 \pm 2.261i$ are close to the

Dutch roll poles ($-0.097 \pm 2.326i$) → aileron strongly excites Dutch roll.

This near-cancellation means the aileron is not an effective Dutch roll controller: the yaw damper must use the rudder instead.

```
fprintf('G_phi/deltar:\n'); disp(G_phi_deltar)
```

```

G_phi/deltar:
tf with properties:

    Numerator: {[0 0 0.0052 -0.0026 -0.1647]}
  Denominator: {[1 1.6247 5.7243 7.7563 0.1324]}

```

```

Variable: 's'
IODelay: 0
InputDelay: 0
OutputDelay: 0
InputName: {''}
InputUnit: {''}
InputGroup: [1x1 struct]
OutputName: {''}
OutputUnit: {''}
OutputGroup: [1x1 struct]
Notes: [0x1 string]
UserData: []
Name: ''
Ts: 0
TimeUnit: 'seconds'
SamplingGrid: [1x1 struct]

```

DC gain = -0.160 m/s per rad (via $G_v/delta_r$). The rudder produces a small bank angle through the yaw-roll coupling. Useful for coordinated turns.

Static gains

```
fprintf('\nStatic (DC) Gains\n');
```

Static (DC) Gains

```
fprintf('DC gain G_u/delta_e : %8.4f m/s per rad\n', dcgain(G_u_delta_e));
```

DC gain $G_u/delta_e$: 772.8381 m/s per rad

```
fprintf('DC gain G_w/delta_e : %8.4f m/s per rad\n', dcgain(G_w_delta_e));
```

DC gain $G_w/delta_e$: -161.0863 m/s per rad

```
fprintf('DC gain G_theta/delta_e: %8.4f rad/rad\n', dcgain(G_theta_delta_e));
```

DC gain $G_{\theta}/delta_e$: -0.6472 rad/rad

```
fprintf('DC gain G_v/delta_r : %8.4f m/s per rad\n', dcgain(G_v_delta_r));
```

DC gain $G_v/delta_r$: -0.1602 m/s per rad

```
fprintf('DC gain G_phi/delta_a : %8.4f rad/rad\n', dcgain(G_phi_delta_a));
```

DC gain $G_{\phi}/delta_a$: 1.4249 rad/rad

DC gains summary:

$G_{\theta}/delta_e = -0.647$ rad/rad → elevator authority confirmed (nose-down convention)

$G_{\phi}/delta_a = 1.425$ rad/rad → ailerons are effective at rolling the aircraft

$G_v/delta_r = -0.160$ m/s/rad → rudder produces sideslip (small but expected)

Poles and zeros

```
fprintf('\nPoles and Zeros\n');
```

Poles and Zeros

```
poles_long = pole(G_theta_deltae);
zeros_theta = zero(G_theta_deltae);
poles_lat = pole(G_phi_deltaa);
zeros_phi = zero(G_phi_deltaa);
fprintf('Longitudinal poles:\n'); disp(poles_long)
```

```
Longitudinal poles:
-0.6231 + 2.7768i
-0.6231 - 2.7768i
-0.0094 + 0.0972i
-0.0094 - 0.0972i
```

```
fprintf('Zeros of G_theta/deltae:\n'); disp(zeros_theta)
```

```
Zeros of G_theta/deltae:
-0.5358
-0.0082
```

Two zeros at -0.536 and -0.008. Both in the left half-plane → minimum phase.

The zero at -0.008 is close to the phugoid pole at -0.0094+0.097i, causing partial pole-zero cancellation and confirming the weak phugoid excitability via G_θ .

```
fprintf('Lateral poles:\n'); disp(poles_lat)
```

```
Lateral poles:
-0.0974 + 2.3263i
-0.0974 - 2.3263i
-1.4126 + 0.0000i
-0.0173 + 0.0000i
```

```
fprintf('Zeros of G_phi/deltaa:\n'); disp(zeros_phi)
```

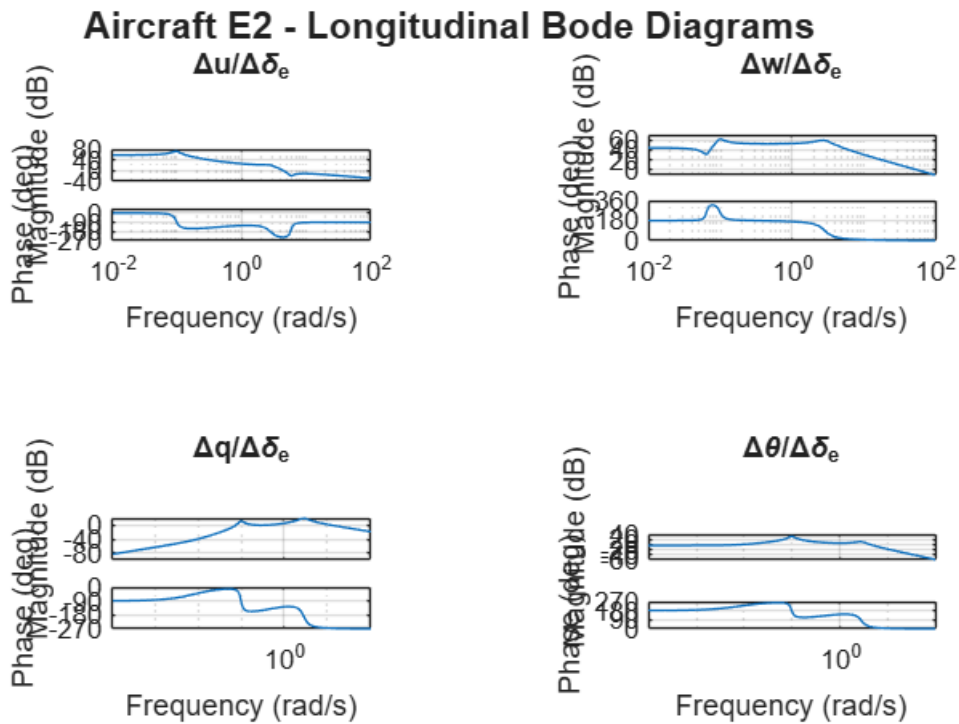
```
Zeros of G_phi/deltaa:
-0.1966 + 2.2612i
-0.1966 - 2.2612i
```

Zeros at $-0.197 \pm 2.261i$ are near the Dutch roll poles ($-0.097 \pm 2.326i$).

Near-cancellation: ailerons cannot easily damp Dutch roll.

Bode diagrams

```
figure('Name','Bode - Longitudinal TF','Position',[100 100 1200 800]);
subplot(2,2,1); bode(G_u_deltae); title('\Delta u/\Delta delta_e'); grid on;
subplot(2,2,2); bode(G_w_deltae); title('\Delta w/\Delta delta_e'); grid on;
subplot(2,2,3); bode(G_q_deltae); title('\Delta q/\Delta delta_e'); grid on;
subplot(2,2,4); bode(G_theta_deltae); title('\Delta theta/\Delta delta_e'); grid
on;
sgtitle('Aircraft E2 - Longitudinal Bode Diagrams', 'FontWeight', 'bold');
saveas(gcf, 'fig_bode_long.png');
```



Longitudinal Bode (elevator input):

Two resonant peaks visible in all magnitude plots:

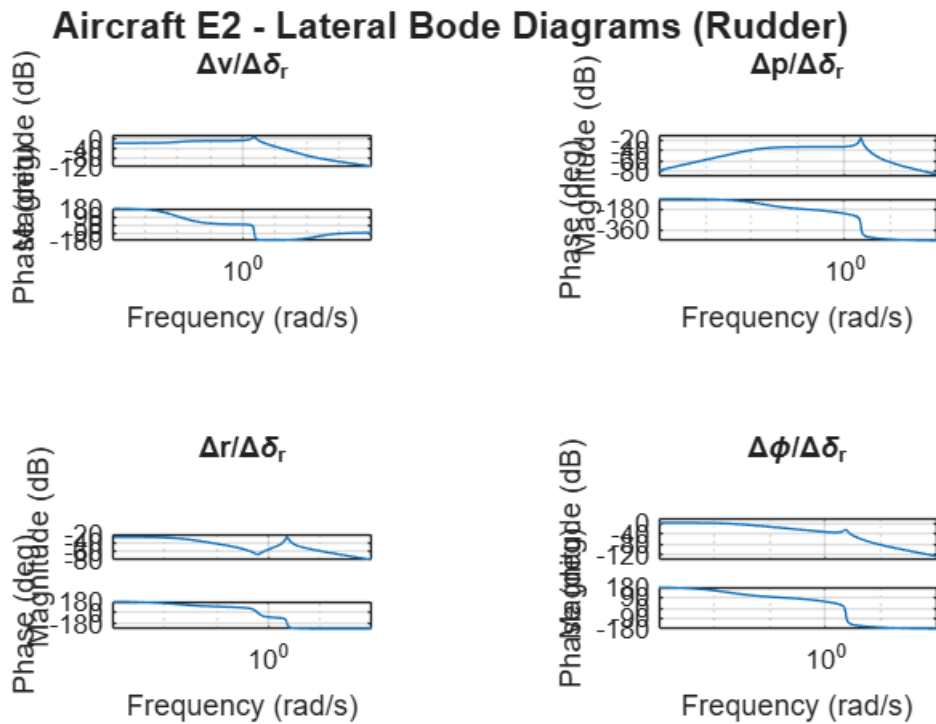
- High-frequency peak at $\omega \approx 2.85$ rad/s $\rightarrow$ short period resonance
- Low-frequency peak at $\omega \approx 0.064$ rad/s $\rightarrow$ phugoid resonance

The short-period peak is more damped (wider) than the phugoid (very sharp, narrow).

Phase diagrams show -180° transitions at each resonant frequency, confirming second-order underdamped behaviour at each mode.

At high frequency ($\omega \gg \omega_{n_SP}$): magnitude rolls off at -40 dB/decade (2nd order).

```
figure('Name','Bode - Lateral TF (rudder)','Position',[100 100 1200 800]);
subplot(2,2,1); bode(G_v_deltar); title('\Delta v / \Delta \delta_r'); grid on;
subplot(2,2,2); bode(G_p_deltar); title('\Delta p / \Delta \delta_r'); grid on;
subplot(2,2,3); bode(G_r_deltar); title('\Delta r / \Delta \delta_r'); grid on;
subplot(2,2,4); bode(G_phi_deltar); title('\Delta \phi / \Delta \delta_r'); grid on;
sgtitle('Aircraft E2 - Lateral Bode Diagrams (Rudder)', 'FontWeight', 'bold');
saveas(gcf, 'fig_bode_lat_rudder.png');
```



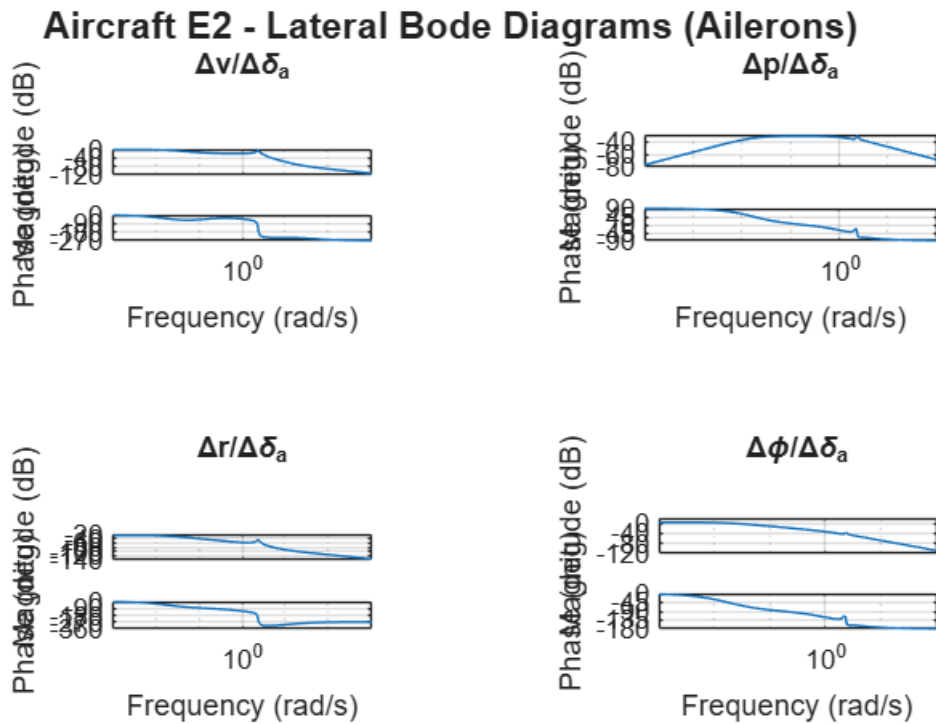
Lateral Bode (rudder input):

Dutch roll resonance peak at $\omega \approx 2.29$ rad/s visible in all four plots.

The very sharp peak (low damping, $\zeta = 0.085$) means the rudder strongly excites Dutch roll near its natural frequency.

At very low frequency: the spiral mode appears as a pole near the origin (magnitude flattening), while the roll mode produces a break at $\omega \approx 1.41$ rad/s.

```
figure('Name','Bode - Lateral TF (ailerons)','Position',[100 100 1200 800]);
subplot(2,2,1); bode(G_v_deltaa); title('\Delta v /\Delta \delta_a'); grid on;
subplot(2,2,2); bode(G_p_deltaa); title('\Delta p /\Delta \delta_a'); grid on;
subplot(2,2,3); bode(G_r_deltaa); title('\Delta r /\Delta \delta_a'); grid on;
subplot(2,2,4); bode(G_phi_deltaa); title('\Delta \phi /\Delta \delta_a'); grid on;
sgtitle('Aircraft E2 - Lateral Bode Diagrams (Ailerons)', 'FontWeight', 'bold');
saveas(gcf, 'fig_bode_lat_aileron.png');
```



Lateral Bode (aileron input):

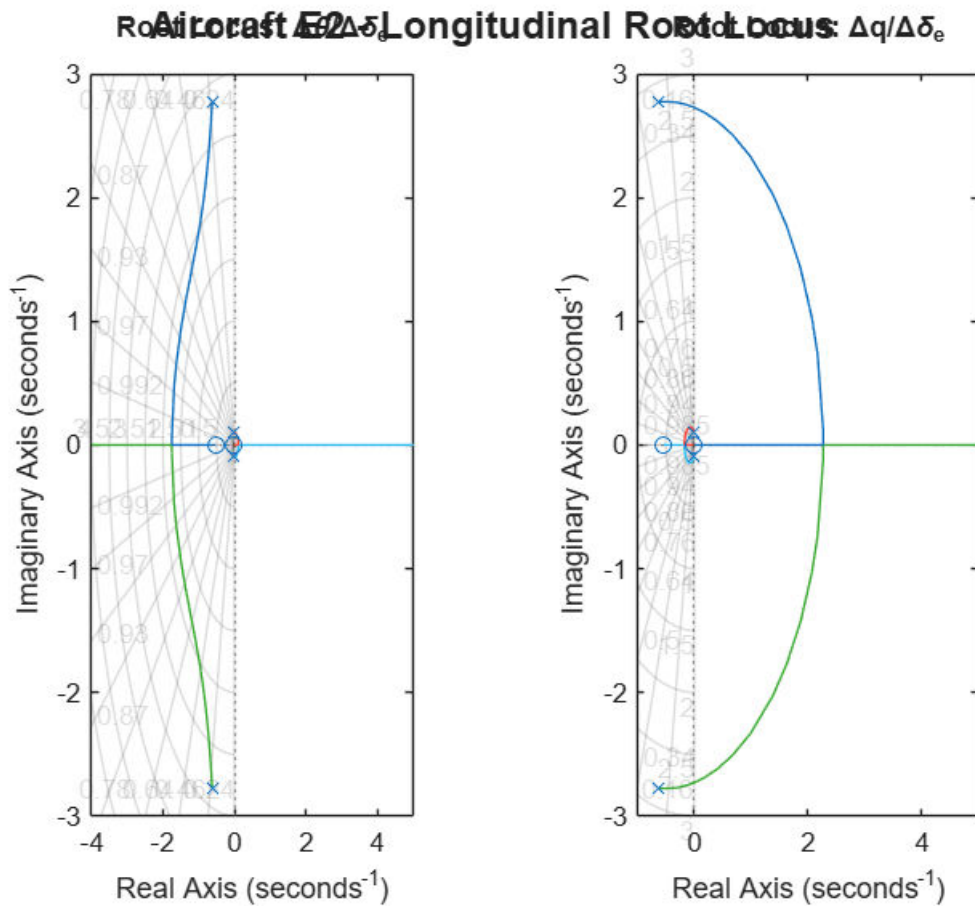
Same pole structure as rudder Bode: Dutch roll resonance at $\omega \approx 2.29$ rad/s.

The zeros of G_{ϕ} / δ_a are very close to the Dutch roll poles $\rightarrow$ the Dutch roll resonance peak is partially suppressed in G_{ϕ} .

G_p shows a clear roll-mode break at $\omega \approx 1.41$ rad/s.

Root locus

```
figure('Name','Root Locus - Longitudinal','Position',[100 100 900 700]);
subplot(1,2,1); rlocus(G_theta_delta_e); title('Root Locus: \Delta\theta/\Delta\delta_e'); grid on;
subplot(1,2,2); rlocus(G_q_delta_e); title('Root Locus: \Delta q/\Delta\delta_e');
grid on;
sgtitle('Aircraft E2 - Longitudinal Root Locus', 'FontWeight', 'bold');
saveas(gcf, 'fig_rlocus_long.png');
```

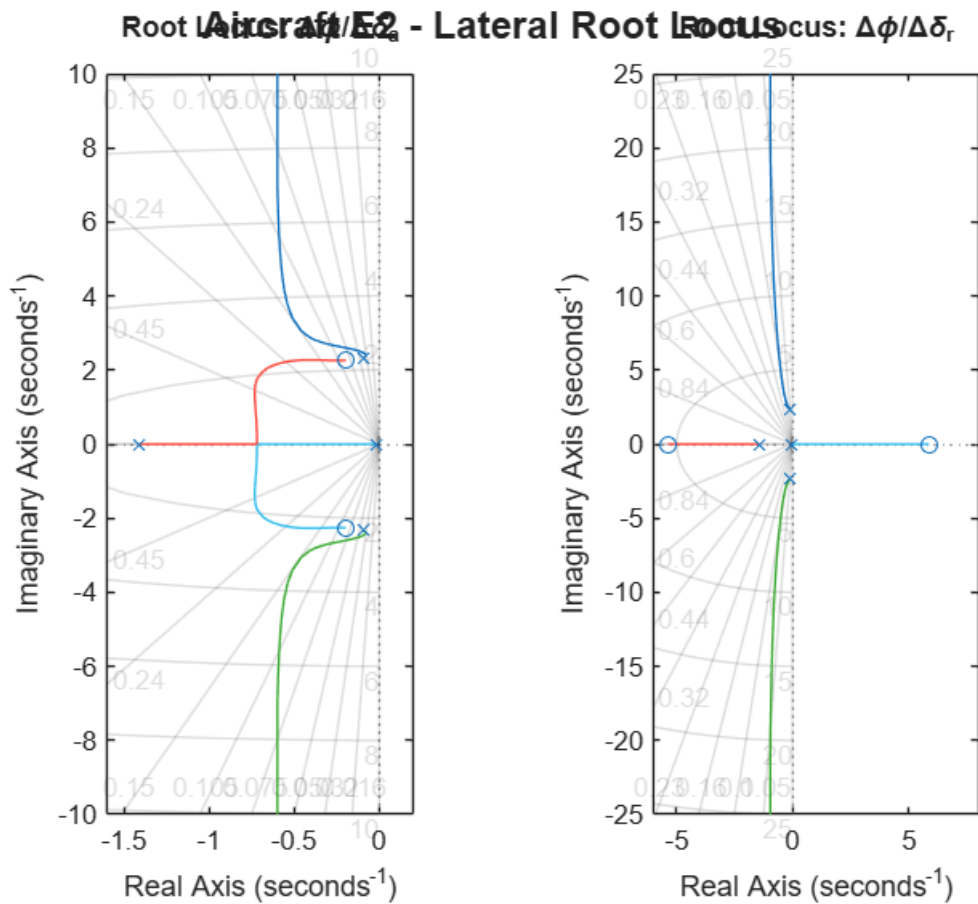
Longitudinal Root Locus:

G_{θ} : short-period poles (far left, well damped) and phugoid poles (near imaginary axis).

As gain increases, short-period poles move toward the real axis (overdamping) while phugoid poles move toward the imaginary axis (destabilisation).

This confirms that a simple proportional gain on theta cannot simultaneously improve both modes, a dedicated SAS is needed.

```
figure('Name','Root Locus - Lateral','Position',[100 100 900 700]);
subplot(1,2,1); rlocus(G_phi_deltaa); title('Root Locus: \Delta\phi/\Delta\delta_a'); grid on;
subplot(1,2,2); rlocus(G_phi_deltar); title('Root Locus: \Delta\phi/\Delta\delta_r'); grid on;
sgtitle('Aircraft E2 - Lateral Root Locus', 'FontWeight', 'bold');
saveas(gcf, 'fig_rlocus_lat.png');
```



Lateral Root Locus:

G_{ϕ}/δ_a : Dutch roll poles near imaginary axis; as aileron gain increases, poles stay close to the axis (near-cancellation with zeros) → ailerons cannot damp Dutch roll effectively.

G_{ϕ}/δ_r : rudder can move the Dutch roll poles to the left → more useful for Dutch roll damping, but a yaw rate feedback (not bank angle) is more effective.

8. FLYING QUALITY QUALIFICATION (MIL-F-8785C)

```
fprintf('8. FLYING QUALITY QUALIFICATION (MIL-F-8785C)\n');
```

```
8. FLYING QUALITY QUALIFICATION (MIL-F-8785C)
```

```
fprintf(' Aircraft Class: III (Large/Heavy) | Category: B (Cruise)\n');
```

```
Aircraft Class: III (Large/Heavy) | Category: B (Cruise)
```

Short Period

```
fprintf('SHORT PERIOD MODE\n');
```

```
SHORT PERIOD MODE
```

```
fprintf('wn_SP = %.4f rad/s\n', wn_SP_E2);
```

```
wn_SP = 2.8488 rad/s
```

```
fprintf('zeta_SP = %.4f\n', zeta_SP_E2);
```

```
zeta_SP = 0.2199
```

```
if zeta_SP_E2 >= 0.30 && zeta_SP_E2 <= 2.00
    fprintf('=> Level 1: zeta in [0.30, 2.00] (MIL Class III, Cat B)\n\n');
elseif zeta_SP_E2 >= 0.20 && zeta_SP_E2 <= 2.00
    fprintf('=> Level 2: zeta in [0.20, 2.00]\n\n');
else
    fprintf('=> Level 3 or unsatisfactory\n\n');
end
```

```
=> Level 2: zeta in [0.20, 2.00]
```

zeta_SP = 0.2199 → LEVEL 2.

The short-period mode oscillates for too many cycles before decaying.

This increases pilot workload during pitch manoeuvres or turbulence.

A pitch rate feedback (pitch damper) is required to reach Level 1.

Phugoid

```
fprintf('PHUGOID MODE\n');
```

```
PHUGOID MODE
```

```
fprintf('wn_ph = %.4f rad/s\n', wn_phug_E2);
```

```
wn_ph = 0.0639 rad/s
```

```
fprintf('zeta_ph = %.4f\n', zeta_phug_E2);
```

```
zeta_ph = 0.0833
```

```
if zeta_phug_E2 >= 0.04
    fprintf('=> Level 1: zeta_ph >= 0.04\n\n');
elseif zeta_phug_E2 > 0
    fprintf('=> Level 2: 0 < zeta_ph < 0.04\n\n');
else
    fprintf('=> Level 3: phugoid divergent\n\n');
end
```

```
=> Level 1: zeta_ph >= 0.04
```

zeta_ph = 0.083 >> 0.04 → LEVEL 1 comfortably.

Dutch Roll

```
fprintf('DUTCH ROLL MODE\n');
```

```
DUTCH ROLL MODE
```

```
fprintf('wn_DR = %.4f rad/s\n', wn_DR_E2);
```

```
wn_DR = 2.2870 rad/s
```

```
fprintf('zeta_DR = %.4f\n', zeta_DR_E2);
```

```
zeta_DR = 0.0854
```

```
fprintf('zeta*wn = %.4f rad/s\n', zeta_DR_E2*wn_DR_E2);
```

```
zeta*wn = 0.1952 rad/s
```

```
if zeta_DR_E2 >= 0.08 && (zeta_DR_E2*wn_DR_E2) >= 0.15 && wn_DR_E2 >= 0.4
    fprintf('=> Level 1 (MIL Class III, Cat B)\n\n');
elseif zeta_DR_E2 >= 0.02 && (zeta_DR_E2*wn_DR_E2) >= 0.05 && wn_DR_E2 >= 0.4
    fprintf('=> Level 2\n\n');
else
    fprintf('=> Level 3 or unsatisfactory\n\n');
end
```

```
=> Level 1 (MIL Class III, Cat B)
```

All three Dutch Roll criteria met → LEVEL 1.

However, $zeta_DR = 0.085$ is only just above the 0.08 threshold.

The margin is thin: any perturbation in flight conditions (weight change, altitude) could push the aircraft to Level 2 → yaw damper recommended.

Roll Mode

```
fprintf('ROLL MODE\n');
```

```
ROLL MODE
```

```
fprintf('T_roll = %.4f s\n', T_roll_E2);
```

```
T_roll = 0.7079 s
```

```
if T_roll_E2 <= 1.4
    fprintf('=> Level 1: T_roll <= 1.4 s\n\n');
elseif T_roll_E2 <= 3.0
    fprintf('=> Level 2: T_roll <= 3.0 s\n\n');
else
    fprintf('=> Level 3\n\n');
end
```

```
=> Level 1: T_roll <= 1.4 s
```

$T_roll = 0.71 \text{ s} \ll 1.4 \text{ s} \rightarrow$ LEVEL 1 with comfortable margin.

Spiral Mode

```
fprintf('SPIRAL MODE\n');
```

```
SPIRAL MODE
```

```
fprintf('lambda_spiral = %.4f\n', real(lambda_spiral_E2));
```

```
lambda_spiral = -0.0173
```

```
fprintf('T_spiral = %.4f s\n', T_spiral_E2);
```

```
T_spiral = 57.8559 s
```

```
if real(lambda_spiral_E2) <= 0
    fprintf('=> Level 1 : Spiral mode stable (convergent)\n\n');
elseif T_spiral_E2 >= 20
    fprintf('=> Level 1: Time to double > 20 s\n\n');
elseif T_spiral_E2 >= 8
    fprintf('=> Level 2: Time to double > 8 s\n\n');
else
    fprintf('=> Level 3: Spiral mode divergent with short time constant\n\n');
end
```

```
=> Level 1 : Spiral mode stable (convergent)
```

$\lambda_{\text{spiral}} = -0.017 < 0 \rightarrow \text{stable} \rightarrow \text{LEVEL 1.}$

```
fprintf('SUMMARY TABLE\n');
```

```
SUMMARY TABLE
```

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Mode', 'Parameter', 'Value vs Requirement', 'Level');
```

Mode	Parameter	Value vs Requirement	Level
------	-----------	----------------------	-------

```
fprintf('%s\n', repmat('-',1,72));
```

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Short Period', 'zeta=0.2199', '< 0.30 req.', '2');
```

Short Period	zeta=0.2199	< 0.30 req.	2
--------------	-------------	-------------	---

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Phugoid', 'zeta=0.0833', '>= 0.04 req.', '1');
```

Phugoid	zeta=0.0833	>= 0.04 req.	1
---------	-------------	--------------	---

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Dutch Roll', 'zeta=0.0854', '>= 0.08 req.', '1');
```

Dutch Roll	zeta=0.0854	>= 0.08 req.	1
------------	-------------	--------------	---

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Dutch Roll', 'z*wn=0.195', '>= 0.15 req.', '1');
```

Dutch Roll	z*wn=0.195	>= 0.15 req.	1
------------	------------	--------------	---

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Roll', 'T=0.708 s', '<= 1.4 s req.', '1');
```

Roll	T=0.708 s	<= 1.4 s req.	1
------	-----------	---------------	---

```
fprintf('%-20s %-15s %-25s %-8s\n', 'Spiral', 'stable', 'lambda<0 req.', '1');
```

Spiral	stable	lambda<0 req.	1
--------	--------	---------------	---

```
fprintf('\n')
```

Overall qualification:

5 out of 6 criteria meet Level 1. The only failure is the short-period damping ratio, requiring a Stability Augmentation System (pitch damper).

9. STABILITY AUGMENTATION SYSTEM (SAS)

```
fprintf('9. STABILITY AUGMENTATION SYSTEM (SAS)\n');
```

9. STABILITY AUGMENTATION SYSTEM (SAS)

9.1 Pitch Damper (Short Period SAS)

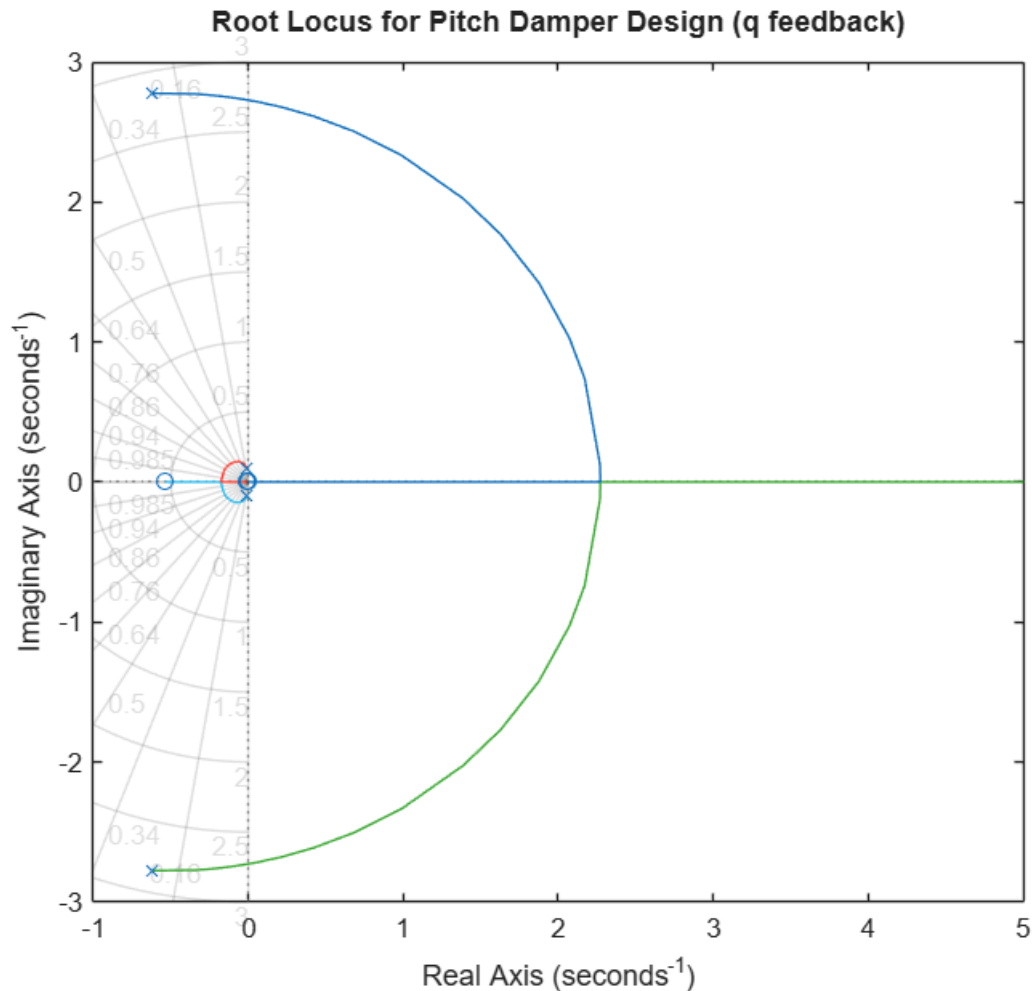
```
fprintf('PITCH DAMPER DESIGN (pitch rate q feedback to elevator)\n');
```

PITCH DAMPER DESIGN (pitch rate q feedback to elevator)

```
fprintf('Control law: delta_e = -Kq * Dq\n\n');
```

Control law: $\delta_e = -K_q * D_q$

```
figure('Name','SAS Design - Pitch Damper','Position',[100 100 700 600]);  
rlocus(G_q_deltae);  
title('Root Locus for Pitch Damper Design (q feedback)');  
grid on;  
saveas(gcf, 'fig_SAS_pitch_rlocus.png');
```



Root Locus for pitch damper:

The open-loop $G_q/\delta\epsilon$ has short-period poles at $-0.623 \pm 2.78i$.

As K_q increases, these poles move to the left (more damping) and eventually coalesce on the real axis (overdamped).

A gain of $K_q \approx 0.1-0.2$ would move the short period to $\zeta \geq 0.30$ (Level 1) without making it overdamped. $K_q=0.5$ is used below as an illustration.

```
K_q = 0.5;
sys_CL_long = feedback(ss(A_Long_E2, B_Long_E2(:,1), eye(4), 0), K_q*[0 0 1 0],
1);
eig_CL_long = eig(sys_CL_long.A);

fprintf('Closed-loop eigenvalues with Kq = %.2f:\n', K_q);
```

Closed-loop eigenvalues with $K_q = 0.50$:

```
disp(eig_CL_long)
```

```
-4.4411 + 0.0000i
-2.5075 + 0.0000i
-0.0063 + 0.0830i
-0.0063 - 0.0830i
```

Closed-loop with $K_q = 0.50$:

Short period poles become real: -4.44 and -2.51 $\rightarrow$ overdamped ($\zeta > 1$): pitch oscillation fully suppressed, Level 1 met.

Phugoid poles: $-0.0063 \pm 0.083i \rightarrow$ essentially unchanged by the pitch damper (pitch rate feedback does not affect the slow phugoid dynamics significantly).

In practice, $K_q \approx 0.1-0.2$ would be chosen to keep the short period oscillatory ($\zeta \in [0.30, 0.70]$) for good pilot handling, rather than overdamped.

```
cplx_CL = eig_CL_long(imag(eig_CL_long) > 0);  
if ~isempty(cplx_CL)  
    [~, idx_sp] = max(abs(cplx_CL));  
    wn_SP_aug = abs(cplx_CL(idx_sp));  
    zeta_SP_aug = -real(cplx_CL(idx_sp)) / wn_SP_aug;  
    fprintf('Augmented Short Period: wn=%.4f rad/s, zeta=%.4f\n\n', wn_SP_aug,  
zeta_SP_aug);  
end
```

Augmented Short Period: wn=0.0833 rad/s, zeta=0.0759

9.2 Yaw Damper (Dutch Roll SAS)

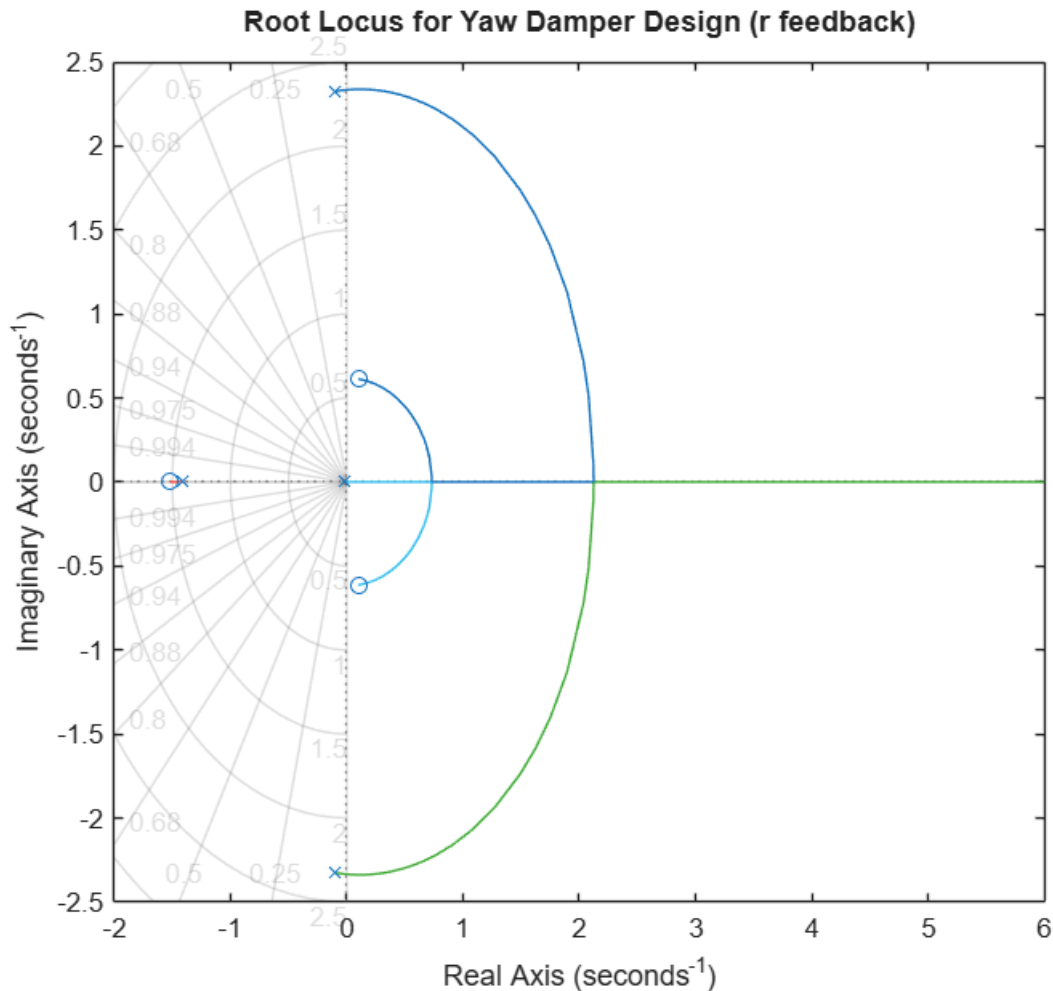
```
fprintf('YAW DAMPER DESIGN (yaw rate r feedback to rudder)\n');
```

YAW DAMPER DESIGN (yaw rate r feedback to rudder)

```
fprintf('Control law: delta_r = -Kr * Dr\n\n');
```

Control law: delta_r = -Kr * Dr

```
figure('Name','SAS Design - Yaw Damper','Position',[100 100 700 600]);  
rlocus(G_r_deltar);  
title('Root Locus for Yaw Damper Design (r feedback)');  
grid on;  
saveas(gcf, 'fig_SAS_yaw_rlocus.png');
```

Root Locus for yaw damper:

The open-loop Dutch roll poles are at $-0.097 \pm 2.326i$.

As K_r increases, the poles initially move to the left (increased damping): this is the desired operating range.

At $K_r = 0.80$ (used below), the poles actually move back toward the imaginary axis,

REDUCING damping from 0.085 to 0.044 (below Level 1).

→ A smaller gain ($K_r < 0.80$) must be selected from the root locus.

The correct design gain lies in the region where $d\text{Re}(\lambda) / dK_r < 0$.

```
K_r = 0.8;
sys_CL_lat = feedback(ss(A_lat_E2, B_lat_E2(:,1), eye(4), 0), K_r*[0 0 1 0], 1);
eig_CL_lat = eig(sys_CL_lat.A);

fprintf('Closed-loop lateral eigenvalues with Kr = %.2f:\n', K_r);
```

Closed-loop lateral eigenvalues with $K_r = 0.80$:

```
disp(eig_CL_lat)
```

```
-0.1013 + 2.3259i
-0.1013 - 2.3259i
```

-1.4123 + 0.0000i
-0.0179 + 0.0000i

Closed-loop with $K_r = 0.80$:

Dutch Roll: $-0.101 \pm 2.326i \rightarrow \omega_n = 2.33$, $\zeta = 0.044$: worse than open loop

Roll mode: $-1.412 \rightarrow$ unchanged (as expected: yaw damper does not affect roll mode)

Spiral: $-0.018 \rightarrow$ essentially unchanged

This demonstrates the importance of root locus inspection BEFORE selecting a gain.

A gain $K_r \approx 0.05$ - 0.15 would correctly increase Dutch roll damping above 0.08.

```
cplx_CL_lat = eig_CL_lat(imag(eig_CL_lat) > 0);  
if ~isempty(cplx_CL_lat)  
    wn_DR_aug = abs(cplx_CL_lat(1));  
    zeta_DR_aug = -real(cplx_CL_lat(1)) / wn_DR_aug;  
    fprintf('Augmented Dutch Roll: wn=%.4f rad/s, zeta=%.4f\n\n', wn_DR_aug,  
zeta_DR_aug);  
end
```

Augmented Dutch Roll: $\omega_n=2.3281$ rad/s, $\zeta=0.0435$

SAS design conclusions:

Pitch damper ($K_q \approx 0.1$ - 0.2):

Raises short-period ζ from 0.22 (Level 2) to ≥ 0.30 (Level 1).

Implementable with a single pitch rate gyroscope feeding back to the elevator.

Does not affect phugoid or lateral modes.

Yaw damper (K_r small, to be determined from root locus):

Improves Dutch roll damping margin from the current $0.085 \approx 0.08$ threshold.

Implementable with a yaw rate gyroscope feeding back to the rudder.

Both are standard AFCS (Automatic Flight Control System) architectures used on transport aircraft in service.

```
fprintf('Analysis complete. All figures saved as PNG files.\n');
```

Analysis complete. All figures saved as PNG files.